Short questions

|!|BCESC0280000001|!|

(a) Describe the major steps of DNA fingerprinting. (4 marks)

(b) Explain how DNA fingerprinting can be used to determine parent-child relationships. (2 marks)

(c) Suggest ***one*** other application of DNA fingerprinting. (1 mark)

##

(a) DNA is extracted from cells. (1)
DNA is cut into DNA fragments using restriction enzymes. / DNA fragments containing highly variable regions are obtained. (1)
The DNA fragments are loaded into an agarose gel slab and placed in an electric field. / The DNA fragments are separated into bands according to their sizes by gel electrophoresis. (1)
The gel is stained with a dye to reveal the DNA bands for comparison. (1)

(b) Children inherit their DNA from their parents. (1)
The DNA bands of a child are either common to those of the mother or those of the father. (1)

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(c) Any ***one*** of the following: (1)

- As a forensic tool for identifying suspects / victims / human remains / missing persons
- Applications in palaeontology / evolutionary studies
- Applications in archaeology
- Applications in medical diagnostics (screening patients and fetuses for inherited diseases)
- Matching organ donors
- Authentication of food products / herbal medicines
- Conservation of endangered species (e.g. by tracing the origin of ivory products)

(Accept other reasonable answers.)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Applying concepts	2
(c)	Understanding basic concepts	1

|!|BCESB0280000002|!| ★ **Cross Topic**

For each of the enzymes listed in ***Column 1***, select from ***Column 2*** one phrase that correctly describes its function. Put the appropriate letter in the space provided.

(3 marks)

Column 1		***Column 2***
DNA polymerase	(a) _____	A. catalyses the formation of mRNA
Restriction enzyme	(b) _____	B. cuts DNA at specific base sequences
DNA ligase	(c) _____	C. catalyses the formation of new DNA strands
		D. joins DNA fragments together

##

(a) C (1)

(b) B (1)

(c) D (1)

(3 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESC0280000003|!|

Complete the following paragraph with the appropriate terms. (5 marks)

(a)_____ refers to the techniques involved in the construction of recombinant DNA molecules, which contain DNA from different sources. Through proper processes, DNA fragments that contain a target gene are obtained. Using a (b)_____, which cuts DNA at a specific base sequence, the target gene is cut out and bacterial (c)_____ are cut open. The two cut molecules can be joined together with the help of a (d)_____. The resulting DNA molecule formed is called a (e)_____.

##

(a) Recombinant DNA technology (1)
(b) restriction enzyme (1)
(c) plasmids (1)
(d) DNA ligase (1)
(e) recombinant plasmid (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESC0280000004|!| **Cross Topic**

Recombinant DNA technology and genetically modified organisms (GMOs) have wide applications in many fields. With recombinant DNA technology, scientists can now transfer a gene for a particular characteristic from one organism to another.

(a) Name ***two*** types of enzymes that are usually used in producing recombinant DNA. Describe their functions. (4 marks)

(b) Why can the genes from another species be expressed in GMOs? (1 mark)

(c) Suggest ***one*** application of recombinant DNA technology in medicine and ***one*** application in agriculture. (2 marks)

##

(a) Restriction enzyme (1)
A restriction enzyme recognizes and cuts DNA at a particular base sequence. (1)
DNA ligase (1)
A DNA ligase joins DNA fragments together. (1)

(b) The genetic code is universal. / The same triplet of DNA bases codes for the same amino acid in all organisms. (1)

(c) Application in medicine: producing human insulin / human growth hormone / vaccines (or other reasonable answers) (1)
Application in agriculture: producing GM crops with pest resistance / with disease resistance / with longer shelf life / with improved nutritional value (or other reasonable answers) (1)

(7 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2

|!|BCESC0280000005|!| **Cross Topic**

Genetic engineering is the direct manipulation of genetic material to alter the characteristics of organisms. This may involve the transfer of genes between organisms of the same or different species so as to cause the development of particular characteristics.

(a) How do genes determine the characteristics of an organism? (1 mark)

(b) State a vector that is commonly used in genetic engineering to transfer genes between organisms. (1 mark)

(c) Suggest ***two*** ways in which genetically modified organisms (GMOs) can be beneficial to humans. (2 marks)

(d) Suggest ***two*** potential hazards of genetic engineering. (2 marks)

##

(a) Genes determine the characteristics of an organism by controlling what proteins are made by the organism. (1)

(b) Bacterial plasmid (1)

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(c) Any ***two*** of the following: (2)

- GMOs can be used to produce pharmaceutical products / human insulin / human growth hormones / vaccines.
- Growing GM crops and animals with higher productivity can increase food supply.
- Growing GM crops with resistance to pest can reduce the use of pesticides / pollution.
- Food produced by GMOs may have a higher nutritional value.
- GM bacteria can be used to clean up oil spills.

(Accept other reasonable answers.)

(d) Any ***two*** of the following: (2)

- GM food may cause unknown harmful effects on human health / may cause allergic reactions in some people.
- GMOs may out-compete wild types and replace them / lead to a decrease in biodiversity / upset the ecological balance.
- Antibiotic resistance genes in GM bacteria may be transferred to pathogenic bacteria, resulting in 'superbugs'.

(Accept other reasonable answers.)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Applying concepts	2
(d)	Applying concepts	2

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